# 循环

## 1. 循环

### 1.1 模块介绍

循环组件用于循环流程，使用者将需要循环处理的流程用循环组件的循环起点和循环终点做圈选，启动循环流程。

## 循环

联动激活

任一激活

信息输入

元素序号

0114f480.index

元素值

0114f480.item

数组长度

0114f480.length

循环单元终点

循环单元起点

模块运行结束

## 1.2 模块参数

| 参数名称 | 类型 | 功能介绍 |
| --- | --- | --- |
| **联动激活** | 布尔型 | 该节点可连接多条布尔型输入，当**上游所有条件均为 Ture 时可激活当前组件执行**，即“全部条件满足时，则执行该模块” |
| **任一激活** | 布尔型 | 该节点可连接多条布尔型输入，**上游任一条件传入为 True 可激活当前组件执行逻辑**，即“任一条件满足，则执行该模块” |
| **信息输入** | 任意类型的数组 | 循环的输入需为数组类型，需要保证上游输入为数组类型 |
| **元素序号** | - | 记录执行循环的第n次，可通过 {{xxxx.index}}将数组的执行次数引用取值 |
| **元素值** | - | 即数组中某一个元素，可通过 {{xxxx.item}}将该元素引用取值 |
| **数组长度** | - | 执行数组的长度，可通过 {{xxxx.length}}将该数组长度引用取值 |
| **循环单元终点** | 布尔型 | 将循环执行最后一个模块的“模块运行结束”连接次节点，完成整个循环的设计 |
| **循环单元起点** | 布尔型 | 用于触发循环开始执行的模块 |
| **模块运行结束** | 布尔型 | 该模块运行结束，则输出值为Ture，通常用于触发下游流程 |

## 1.3 使用示例

### 1.3.1 Agent规划

**以下是工作流规划数据，复制后导入到你自己的Agent做测试吧！**

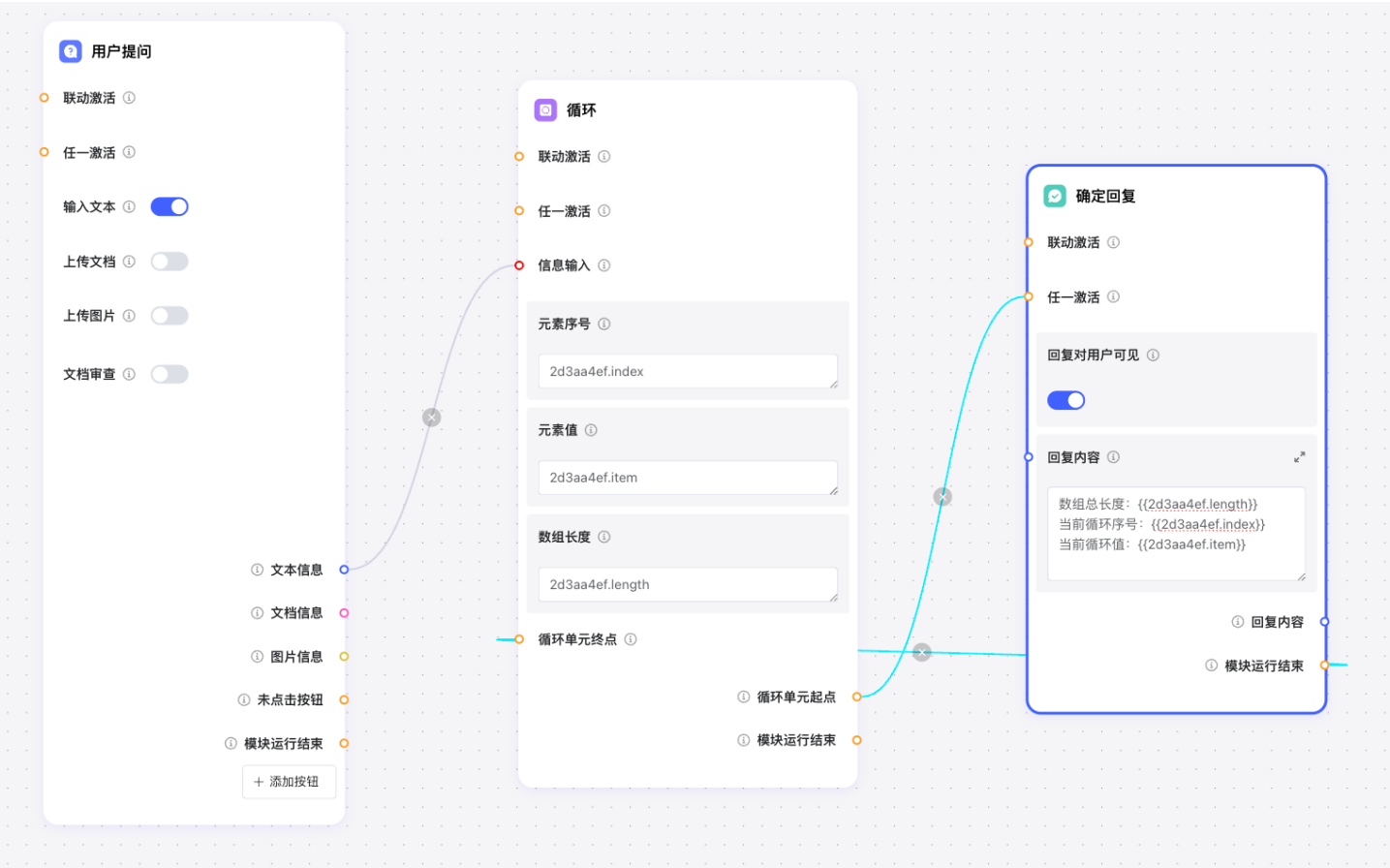
代码块

```
{"nodes":[{"id":"simpleInputId","type":"custom","initialized":false,"position":{"x":-362.3383720930233,"y":367.4154069767442},"data":{"outputs":[{"valueType":"string","description":"引用变量: {{userChatInput}}","label":"文本信息","type":"source","targets":[{"targetHandle":"items","target":"b7658407-8135-40a6-bb78-0eaa692a9fd4"}],"key":"userChatInput"},{"valueType":"file","description":"以JSON数组格式输出用户上传文档列表，若为文档比对，包含分组信息","label":"文档信息","type":"source","targets":[],"key":"files"},{"valueType":"image","description":"以JSON数组格式输出用户上传的图片列表","label":"图片信息","type":"source","targets":[],"key":"images"},{"valueType":"boolean","description":"当未点击任何按钮时值为true","label":"未点击按钮","type":"source","targets":[],"key":"unclickedButton"},
```

{"valueType":"boolean","description":"运行完成后开关打开,下游链接组件开始运
行。","label":"模块运行结束","type":"source","targets":
[],"key":"finish"}],"moduleType":"questionInput","inputs":
[{"connected":true,"valueType":"boolean","description":"同时满足上游所有条件方可激
活当前组件执行逻辑","label":"联动激
活","type":"target","keyType":"trigger","value":false,"key":"switch"},
{"connected":true,"valueType":"boolean","description":"同时满足上游任一条件即可激活
当前组件执行逻辑","label":"任一激
活","type":"target","keyType":"triggerAny","value":false,"key":"switchAny"},
{"valueType":"boolean","description":"输入文本开关","label":"输入文
本","type":"switch","value":true,"key":"inputText"},
{"valueType":"boolean","description":"上传文档开关","label":"上传文
档","type":"switch","value":false,"key":"uploadFile"},
{"valueType":"boolean","description":"上传图片开关","label":"上传图
片","type":"switch","value":false,"key":"uploadPicture"},
{"valueType":"boolean","description":"文档审查开关","label":"文档审
查","type":"switch","value":false,"key":"fileUpload"},
{"valueType":"boolean","description":"是否开启文档比对功能","label":"是否文档对
比","type":"checkBox","value":false,"key":"fileContrast"},
{"valueType":"any","description":"上传的文件列表,如果开启了文档对比,每个分组只能上传
一个文件","label":"文档分组","type":"table","value":[],"key":"fileInfo"},
{"valueType":"boolean","description":"是否作为初始全局input","label":"是否作为初始
全局input","type":"hidden","value":true,"key":"initialInput"}],"intro":"用户输入
入口,对话中用户的输入信息,与其他模块连接,一般作为起始模块","name":"用户提
问","disabled":false,"category":"用户提问"}},{"id":"cf2596c1-6890-45cf-be37-
6b8ba0349332","type":"custom","initialized":false,"position":
{"x":679.2497093023258,"y":518.6029069767442},"data":{"outputs":
[{"valueType":"string","description":"回复内容原样输出。","label":"回复内
容","type":"source","value":"","targets":[],"key":"text"},
{"valueType":"boolean","description":"运行完成后开关打开，下游链接组件开始运
行。","label":"模块运行结束","type":"source","targets":
[{"targetHandle":"loopEnd","target":"b7658407-8135-40a6-bb78-
0eaa692a9fd4"}],"key":"finish"}],"moduleType":"confirmreply","inputs":
[{"connected":true,"valueType":"boolean","description":"同时满足上游所有条件方可激
活当前组件执行逻辑","label":"联动激
活","type":"target","keyType":"trigger","value":false,"key":"switch"},
{"connected":true,"valueType":"boolean","description":"同时满足上游任一条件即可激活
当前组件执行逻辑","label":"任一激
活","type":"target","keyType":"triggerAny","value":false,"key":"switchAny"},
{"connected":false,"valueType":"boolean","description":"控制回复内容是否输出给用
户","label":"回复对用户可见","type":"switch","value":true,"key":"stream"},
{"connected":true,"valueType":"string","description":"可以使用 \\n 来实现连续换
行。\n\n可以通过外部模块输入实现回复，外部模块输入时会覆盖当前填写的内容。引用变量:
{{text}}","label":"回复内容","type":"textarea","value":"数组总长度:
{{2d3aa4ef.length}}\n当前循环序号: {{2d3aa4ef.index}}\n当前循环值:
{{2d3aa4ef.item}}","key":"text"}],"intro":"结合触发条件使用，输出预设内容或输出上游
模块接入内容。","name":"确定回复","disabled":false,"category":"大模型"}},

{"id":"b7658407-8135-40a6-bb78-0eaa692a9fd4","type":"custom","initialized":false,"position":{"x":141.33691860465103,"y":429.6404069767441},"data":{"outputs":[{"valueType":"boolean","description":"循环单元起点","label":"循环单元起点","type":"source","targets":[{"targetHandle":"switchAny","target":"cf2596c1-6890-45cf-be37-6b8ba0349332"}],"key":"loopStart"},{"valueType":"boolean","description":"运行完成后开关打开,下游链接组件开始运行。","label":"模块运行结束","type":"source","targets":[],"key":"finish"}],"moduleType":"forEach","inputs":[{"connected":true,"valueType":"boolean","description":"同时满足上游所有条件方可激活当前组件执行逻辑","label":"联动激活","type":"target","keyType":"trigger","value":false,"key":"switch"},{"connected":true,"valueType":"boolean","description":"同时满足上游任一条件即可激活当前组件执行逻辑","label":"任一激活","type":"target","keyType":"triggerAny","value":false,"key":"switchAny"},{"connected":true,"valueType":"any","description":"JSON数组或者目标对象","label":"信息输入","type":"target","value":"","key":"items"},{"connected":false,"valueType":"number","description":"记录执行循环的次数","label":"元素序号","type":"loopMemoryVar","value":"2d3aa4ef.index","key":"index"},{"connected":false,"valueType":"string","description":"记录执行数组的内容","label":"元素值","type":"loopMemoryVar","value":"2d3aa4ef.item","key":"item"},{"connected":false,"valueType":"number","description":"执行数组的长度","label":"数组长度","type":"loopMemoryVar","value":"2d3aa4ef.length","key":"length"},{"connected":true,"valueType":"boolean","description":"循环单元终点","label":"循环单元终点","type":"target","key":"loopEnd"}],"intro":"依次读取输入数组中的元素，执行循环流程","name":"循环","disabled":false,"category":"高阶能力"}}],"edges":[{"id":"vueflow__edge-simpleInputIduserChatInput-b7658407-8135-40a6-bb78-0eaa692a9fd4items","type":"custom","source":"simpleInputId","target":"b7658407-8135-40a6-bb78-0eaa692a9fd4","sourceHandle":"userChatInput","targetHandle":"items","data":{},"label":"","sourceX":-38.33837209302328,"sourceY":948.3997819767442,"targetY":626.132563959166,"targetX":137.33691860465103,"animated":false},{"id":"vueflow__edge-b7658407-8135-40a6-bb78-0eaa692a9fd4loopStart-cf2596c1-6890-45cf-be37-6b8ba0349332switchAny","type":"custom","source":"b7658407-8135-40a6-bb78-0eaa692a9fd4","target":"cf2596c1-6890-45cf-be37-6b8ba0349332","sourceHandle":"loopStart","targetHandle":"switchAny","data":{},"label":"","sourceX":505.33691860465103,"sourceY":1083.218501459166,"targetY":659.29828311852,"targetX":677.2496511999901,"animated":false},{"id":"vueflow__edge-cf2596c1-6890-45cf-be37-6b8ba0349332finish-b7658407-8135-40a6-bb78-0eaa692a9fd4loopEnd","type":"custom","source":"cf2596c1-6890-45cf-be37-6b8ba0349332","target":"b7658407-8135-40a6-bb78-0eaa692a9fd4","sourceHandle":"finish","targetHandle":"loopEnd","data":{},"label":"","sourceX":1001.2498043962692,"sourceY":1049.7905272273724,"targetY":1022.320063959166,"targetX":137.33691860465103,"animated":false}],"position"

:[428.6031814077639,-89.07845034586822],"zoom":0.5160149169302679,"viewport":{"x":428.6031814077639,"y":-89.07845034586822,"zoom":0.5160149169302679}}




## 1.3.2 输入示例

代码块

```
1   [1,2,3,4,5,6]
```

## 1.3.3 输出结果示例

配置 规划 删除

预览

你

[1,2,3,4,5,6]

test-foreach

数组总长度：6
当前循环序号：1
当前循环值：1

test-foreach

数组总长度：6
当前循环序号：2
当前循环值：2